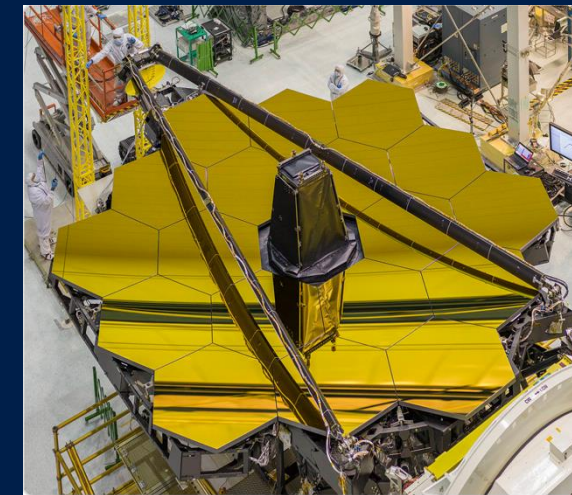


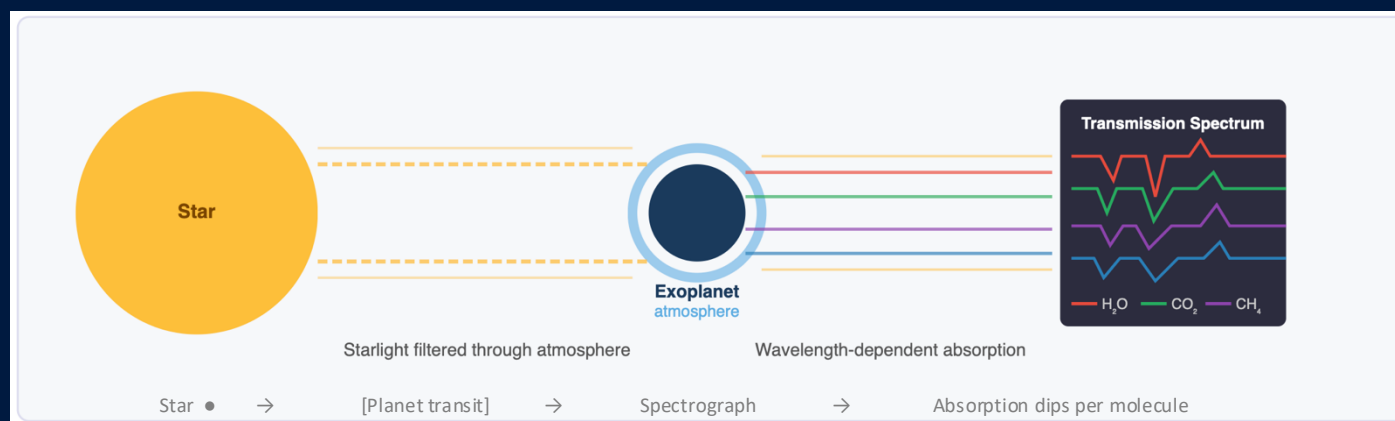
Exploring ML Models for Detection of Atmospheric Composition in Exoplanets



Dataset: INARA (NASA FDL / Zorzan et al. 2025)

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Motivation - Why This Matters

The intersection of scientific urgency and methodological opportunity

The Science

- ▶ Rocky exoplanet atmospheres reveal if a world could be habitable
- ▶ Co-detection of $O_2 + CH_4$ is a strong biosignature- signals biological activity
- ▶ JWST + planned Habitable Worlds Observatory (HWO) produce spectra at unprecedented scale
- ▶ Population-level surveys require analysing thousands of planets simultaneously

Traditional Bayesian Retrieval

- X Nested sampling / MCMC — rigorous but slow
- X Hours to days per planet
- X Does not scale to thousands of targets

ML-Based Retrieval (This project)

- Train once on 124K synthetic profiles -> predict in milliseconds
- INARA: first large-scale ML benchmark (3.1M sims, Zorzan 2025)
- Random Forest baseline: PCA(300) + 12 independent regressors
- 1D CNN: shared conv backbone + 12 per-molecule output heads
- HPC (Northeastern Discovery) used for full 124K production run

```
lnara_project/
├── src/                                # Core ML modules (do not modify)
│   ├── data_utils.py                  # Loading, splitting, normalisation, PCA, metrics
│   ├── baseline_model.py              # Per-molecule Random Forest model
│   └── deep_model.py                  # CNN1D + Trainer + SpectralDataset
├── pipeline/
│   ├── config.yaml                    # All configuration (paths, hyperparameters, toggles)
│   └── steps/                          # Numbered pipeline steps — run in order
│       ├── config_loader.py           # Shared config reader (used by all steps)
│       ├── 01_extract.py              # Step 1: raw archives → processed numpy arrays
│       ├── 02_feature_engineer.py     # Step 2: split + normalise + PCA → artifacts
│       ├── 03_train_baseline.py       # Step 3: Random Forest (≤10k samples)
│       ├── 04_train_deep.py           # Step 4: 1D CNN training (with target normalisation)
│       └── 05_evaluate.py             # Step 5: unified test eval + comparison report
├── jobs/
│   ├── slurm/                          # Northeastern Explorer HPC job scripts
│   │   ├── 01_extract.sh
│   │   ├── 02_feature_engineer.sh
│   │   ├── 03_train_baseline.sh
│   │   ├── 04_train_deep.sh
│   │   └── 05_evaluate.sh
│   └── submit_pipeline.sh             # Submit all jobs with dependency chain
├── local/
│   └── run_pipeline.sh                # Mac M5 — runs all steps sequentially
├── notebooks/
│   └── eda.ipynb                      # Exploratory Data Analysis (standalone)
├── inara_data/
│   ├── processed/                     # Output of Step 1 (spectra.npy etc.)
│   └── engineered/                    # Output of Step 2 (normalised arrays + PCA)
├── models/                            # Saved model weights
│   ├── baseline_rf.joblib
│   └── cnn1d.pt
├── results/                           # Metrics, predictions, training history
├── dashboard.py                       # Streamlit interactive results dashboard
├── process_inara.py                   # Standalone data extraction script
├── run_baseline.py                    # Standalone baseline training script
└── run_deep_model.py                  # Standalone deep model training script
```

Problem Statement

Multi-output regression over molecular $\log_{10}$ abundances

Given a transmission spectrum of an exoplanet's atmosphere,
predict the $\log_{10}$ surface volume mixing ratios of 12 molecular species.

INPUT

Transmission Spectrum

12 CLIMA channels
101 altitude levels
Z-score normalised
(INARA ATMOS format)



ML MODEL

Random Forest

(baseline, $\leq 10k$ samples)

1D CNN

(deep model, full dataset)



OUTPUT

12 Molecular Abundances

$\text{H}_2\text{O} \cdot \text{CO}_2 \cdot \text{O}_2 \cdot \text{O}_3$
 $\text{CH}_4 \cdot \text{N}_2 \cdot \text{N}_2\text{O} \cdot \text{CO}$
 $\text{H}_2 \cdot \text{H}_2\text{S} \cdot \text{SO}_2 \cdot \text{NH}_3$
($\log_{10}$ VMR at surface)

Why $\log_{10}$ space? Mixing ratios span ~ 12 orders of magnitude (~ 0.1 for N_2 to 10^{-40} for trace species). $\log_{10}$ makes the prediction scale uniform — an error of 1.0 means 'off by one order of magnitude'.

Dataset - INARA (NASA FDL / Zorzan et al. 2025)

3.1 million synthetic rocky planet spectra from NASA Planetary Spectrum Generator

Dataset Facts

Full dataset size	3,112,620 synthetic CLIMA profiles
This project uses	124,000 samples
Profile type	INARA ATMOS — CLIMA atmospheric profiles
Altitude levels	101 per profile
Input channels	12 CLIMA variables (T, P, 10 molecules)
Stellar types	F, G, K, M — 4 types
Format	CSV per planet → .npy arrays after processing

CLIMA Input Channels (12)

T / P Temperature & Pressure profiles
— primary physical state of the atmosphere

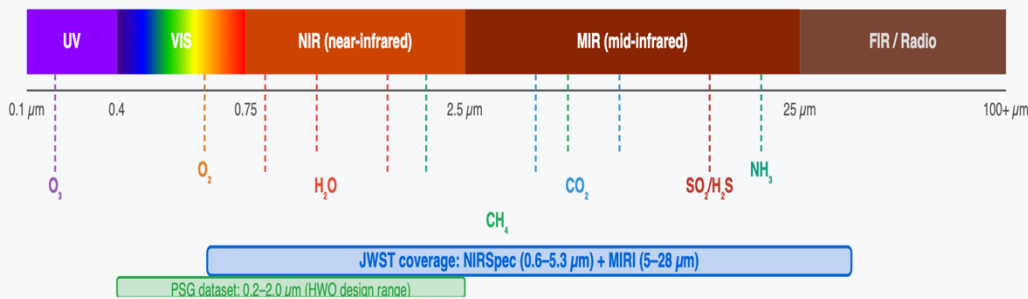
10 mol H₂O, CO₂, O₂, O₃, CH₄, N₂, N₂O, CO, H₂, H₂S
— altitude-resolved mixing ratios

Norm. All 12 channels Z-score normalised per channel (fit on train only)

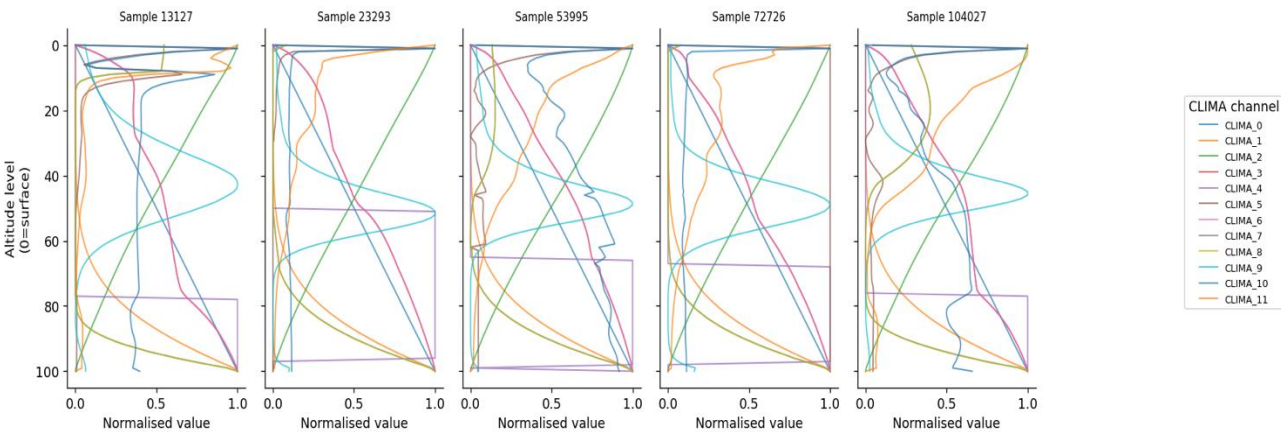
12 Target Molecules

Habitability:	H ₂ O · CO ₂
Biosignatures:	O ₂ · O ₃ · CH ₄ · N ₂ O
Background:	N ₂ · H ₂
Geochemical:	CO · H ₂ S · SO ₂ · NH ₃

Electromagnetic Spectrum — Atmospheric Science Relevant Ranges

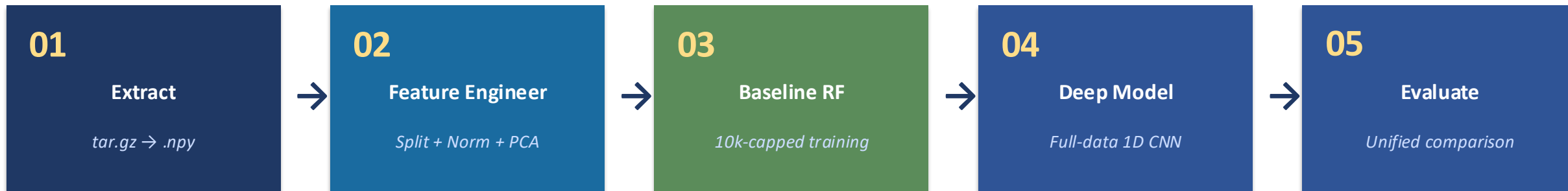


5 Example Atmospheric Profiles (channels normalised per sample)



ML Pipeline & Feature Engineering

Five-stage structured pipeline — each step produces reusable artifacts



Step 2 Deep Dive — Feature Engineering

70 / 15 / 15 Split

- Deterministic (seed=42)
- Fixed once — both models
- use identical test set
- Saved as `train/val/test_indices.npy`

Z-Score Normalisation

- Per CLIMA channel (12 channels)
- Fit on train set ONLY
- Prevents data leakage
- Saved as `scaler.joblib`

PCA (RF only)

- Flatten: $12 \times 101 \rightarrow 1212$ features
- Reduce to 300 components
- Retains 100% variance (25k dataset rank-limited)
- Saved as `pca.joblib`

Deep Model Input

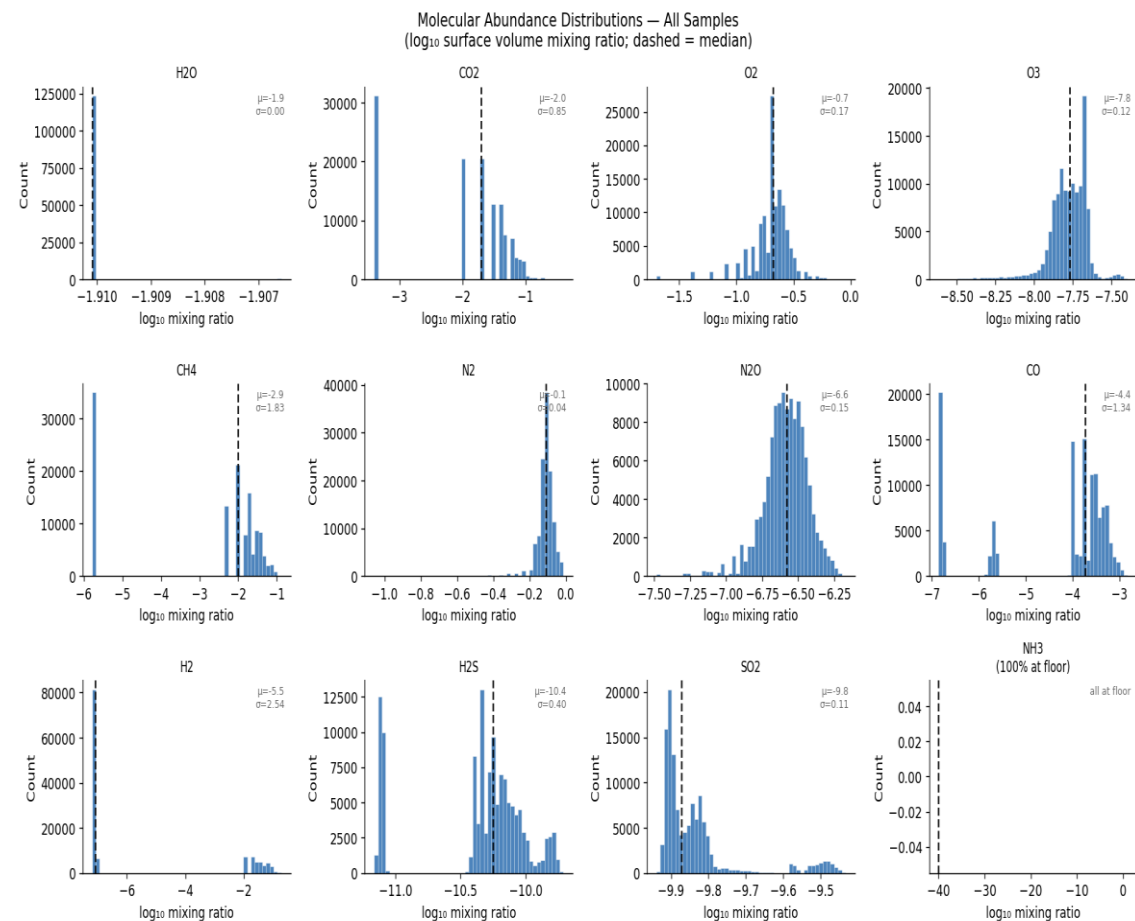
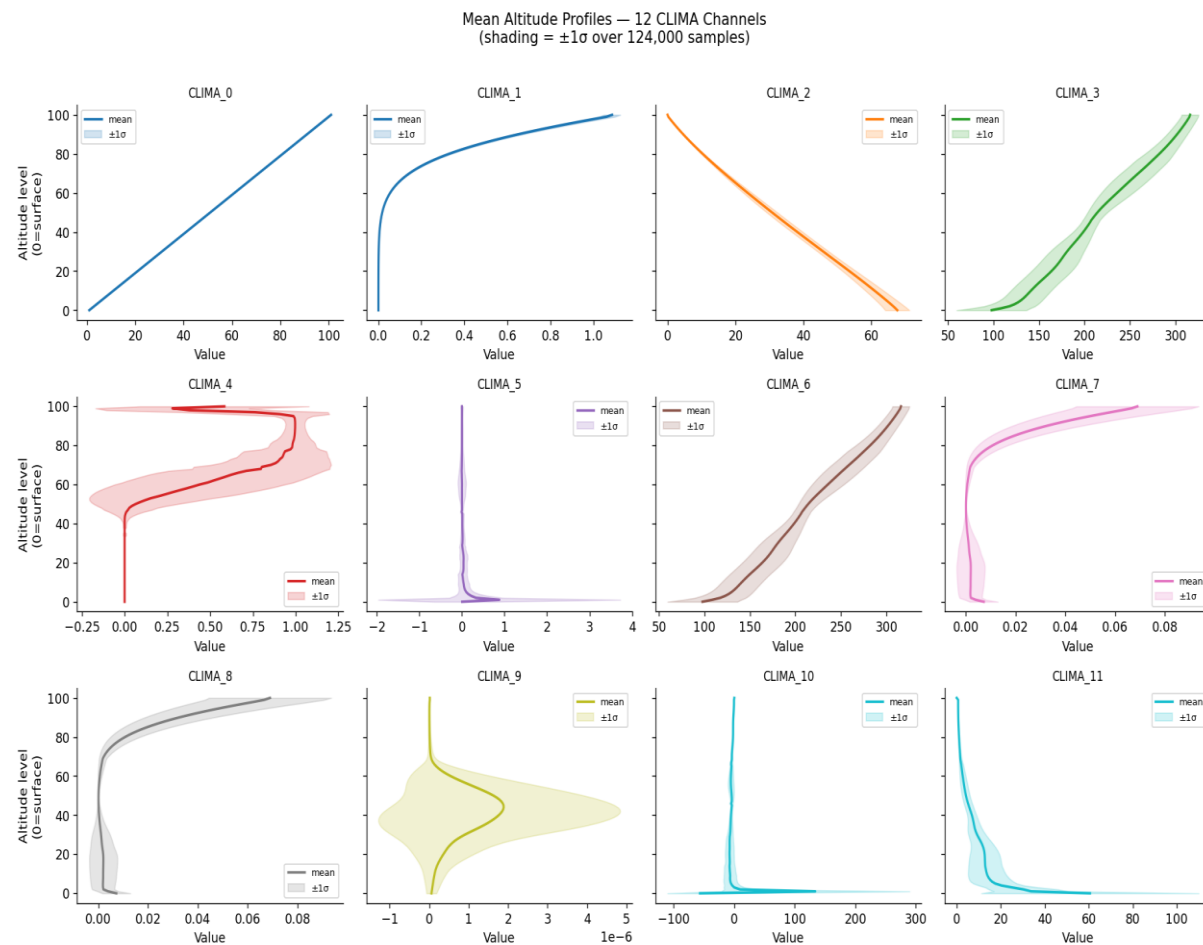
- Raw 2D tensor (12×101)
- No PCA — CNN learns features
- Z-normalised spectra only
- Stored as `spectra_train.npy`

Why Minimal Feature Engineering?

- CLIMA profiles are already physically structured (not raw pixels/text)
- 12 meaningful channels x 101 altitude levels -- no noise features
- Preprocessing: Z-score per channel (fit on train, applied to val/test)
- Log10 targets handle 12-order-of-magnitude range (not FE)
- RF adds PCA(300): dimensionality reduction, retains $\geq 95\%$ variance
- CNN learns altitude patterns internally via stride-2 convolutions

EDA -- Atmospheric Profiles & Molecular Targets

Mean CLIMA altitude profiles (left) | Log10 VMR target distributions (right)



Baseline Model — Random Forest

12 independent RandomForestRegressors on PCA-reduced features ($\leq 10,000$ training samples)

Architecture

Input	(12, 101) CLIMA tensor -- flattened to (1212,)
Dimensionality	PCA(n_components=300, whitening=False)
PCA variance	$\geq 95\%$ total variance retained
Per-molecule	12 independent RandomForestRegressors
n_estimators	100 trees per molecule (1200 trees total)
max_depth	8 (prevents overfitting on 10K sample budget)
Training samples	$\leq 10,000$ per molecule (memory constraint)
Leaves -- CH4	71-169 per tree; median ≈ 110 leaves
NH3 trees	Depth=0 (constant -40) -- degenerate target
Inference speed	~ 10 ms per planet (all 12 molecules)
Interpretability	feature_importance_: top PCA components identified

Why 10k cap?

RF is the fast baseline — < 1 min training, no GPU needed. Fair comparison point.

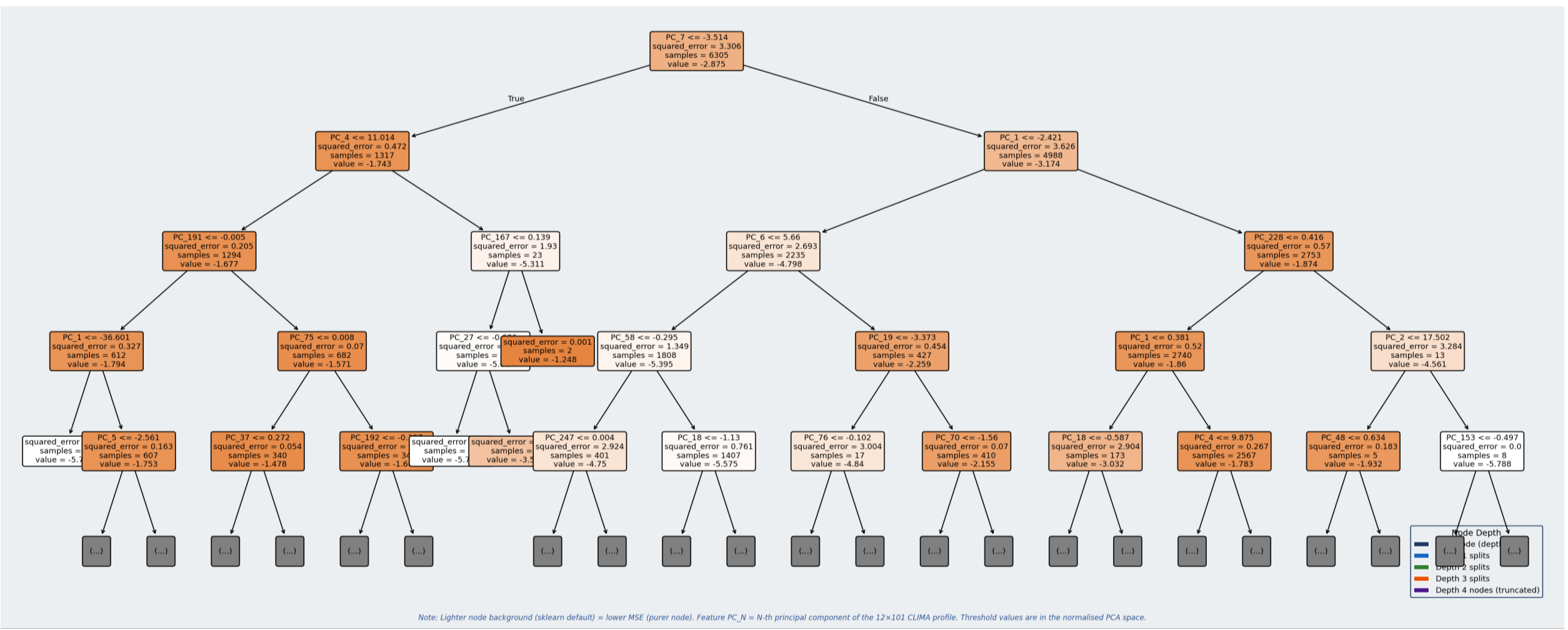
Why RF as Baseline?

- Strong non-parametric regressor -- no gradient tuning required
- PCA handles collinearity across 1212 flattened features
- Interpretable: feature_importance_ reveals top PCA components
- 10K sample cap -> controlled comparison with limited compute
- One model per molecule prevents cross-molecule interference

RF Inference Pipeline

- (1) (12,101) -> flatten -> Z-score normalise -> (1212,)
- (2) PCA.transform: (1212,) -> (300 components)
- (3) For each molecule: RF.predict(x_pca) -> log10 VMR scalar
- (4) Stack 12 scalars -> (12,) output vector
- (5) MoleculeScaler.inverse_transform -> physical log10 values

Baseline Model - Random Forest



Deep Model - Key Design Choices

Every design decision is motivated by the data's properties

1D Convolutions

CLIMA profiles are 1D sequences over altitude. 1D CNNs detect local patterns (e.g. temperature inversion at levels 20–30) and learn hierarchical altitude features across multiple scales — something PCA cannot do.

Shared Backbone

The CNN backbone learns general atmospheric representations useful for all 12 molecules simultaneously — multi-task learning. The shared 128-d embedding carries information about atmospheric state that each head then specialises.

Weighted MSE Loss

Standard MSE would let easy molecules dominate. Weights (1.0–2.0) upweight trace species (SO_2 , NH_3 at 2.0). Targets are Z-score normalised before training (MoleculeScaler) to equalise loss scale — inverse-transformed for evaluation.

Progressive Downsampling

Four conv blocks with decreasing kernel sizes ($9 \rightarrow 7 \rightarrow 5 \rightarrow 3$) and stride-2 + MaxPool compress 101 altitude levels down to 1. Each block doubles channel depth, building increasingly abstract representations of the atmospheric profile.

Per-Molecule Heads

Each molecule has its own MLP head with tuned depth and dropout. Trace species (H_2S , SO_2 , NH_3) get higher dropout (0.25) to prevent overfitting. Stable species (CO_2 , O_2) get lighter heads with lower dropout (0.15).

Gaussian Noise Augmentation

Adding noise $\text{std}=0.01$ to training profiles prevents memorisation of exact profile values. Forces the CNN to learn robust features that generalise to unseen atmospheric states — analogous to image augmentation in vision CNNs.

Deep Model - 1D CNN Architecture

Sequential convolutional backbone + 12 per-molecule output heads | 368,588 parameters

Layer	Output shape	Description
Input	(B, 12, 101)	Normalised CLIMA profile — 12 channels × 101 altitude levels
Conv Block 1	(B, 32, 25)	Conv1d(12→32, k=9, s=2, p=4) + BatchNorm + ReLU + MaxPool1d(2) Conv Block:
Conv Block 2	(B, 64, 6)	Conv1d(32→64, k=7, s=2, p=3) + BatchNorm + ReLU + MaxPool1d(2) Conv1d (k,s,p) + BatchNorm1d
Conv Block 3	(B, 128, 1)	Conv1d(64→128, k=5, s=2, p=2) + BatchNorm + ReLU + MaxPool1d(2) + ReLU + MaxPool1d(2)
Conv Block 4	(B, 256, 1)	Conv1d(128→256, k=3, s=1, p=1) + BatchNorm + ReLU (no MaxPool) [Block 4: no MaxPool]
GlobalAvgPool	(B, 256)	AdaptiveAvgPool1d(1) + Flatten — squeeze altitude dimension
Shared FC	(B, 128)	Dropout(0.25) + FC(256→128) + LayerNorm + ReLU
12 × Head	(B, 12)	Per-molecule MLP: FC(128→hidden→1) + LN + ReLU + Dropout → scalar log ₁₀

Training Strategy

AdamW · Cosine LR schedule · Early stopping · Gradient clipping

Training Configuration

Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Scheduler	CosineAnnealingLR
T_max / eta_min	150 / 1×10^{-6}
Max epochs	150
Early stop patience	30 epochs
Min delta	1×10^{-5}
Batch size	32 (train) / 64 (eval)
Grad clip norm	5.0
Noise augmentation	std = 0.01
Dataset (train)	18,648 samples (124k run)
Target normalisation	Z-score per molecule (MoleculeScaler)

AdamW

Adam with decoupled weight decay — L2 regularisation is applied directly to weights, independent of the adaptive gradient. Better generalisation than standard Adam.

Cosine Annealing

LR starts at 1e-3, smoothly decays to 1e-6 following a cosine curve. No sharp drops. Naturally slows near minima to avoid oscillation.

Early Stopping

Monitors val_loss. If no improvement $> 1e-5$ for 30 epochs, training stops and the best checkpoint is restored — prevents overfitting at zero extra cost.

Gradient Clipping

If gradient norm > 5.0 , all gradients are scaled down proportionally. Prevents exploding gradient spikes that destabilise early training.

Evaluation Methodology

Held-out test set · Per-molecule R^2 , RMSE, MAE · Unified comparison report

R^2 Coefficient of Determination

$$1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2}$$

Primary metric. Scale-normalised — 0.75 means '75% of variance explained' regardless of molecule. Enables fair cross-molecule comparison. $R^2 < 0$ means worse than predicting the mean.

RMSE Root Mean Squared Error

$$\sqrt{\sum (y - \hat{y})^2 / N}$$

In $\log_{10}$ units (orders of magnitude). Penalises large errors more than MAE. A RMSE of 0.5 means predictions are typically $\frac{1}{2}$ an order of magnitude off.

MAE Mean Absolute Error

$$\sum |y - \hat{y}| / N$$

Also in $\log_{10}$ units. More robust to outliers than RMSE. Complementary to RMSE — if $\text{RMSE} \gg \text{MAE}$, there are a few large prediction errors dominating.

Evaluation Fairness Principles

Same test set:

Both models evaluated on the identical 18,468-sample test split, fixed in Step 2 with seed=42. Scores are directly comparable.

Per-molecule granularity:

Reporting R^2 /RMSE/MAE for each of 12 molecules reveals which species each model handles well or poorly.

Intentionally asymmetric training:

RF uses 10k samples; CNN uses 124k. The comparison is baseline vs deep model — not equal-data ablation.

Mean R^2 summary:

Unweighted mean across 12 molecules provides a single-number comparison. NH_3 and N_2 drag it down for both models.

Results — Per-Molecule Test R²

Test 18,648 out of 124k samples - HPC Evaluation

Molecule	RF R ²	RF RMSE	RF MAE	CNN1D R ²	CNN1D RMSE	CNN1D MAE	ΔR ²
H ₂ O †	0.6139	0.000102	0.0000	0.9788	0.0000240	0.0000	+0.365
CO ₂	0.8884	0.2822	0.2084	0.9998	0.0120	0.0076	+0.111
O ₂	0.7285	0.0904	0.0616	0.9987	0.0061	0.0043	+0.270
O ₃	0.7520	0.0624	0.0430	0.9990	0.0040	0.0020	+0.247
CH ₄	0.9276	0.4932	0.3832	0.9999	0.0089	0.0063	+0.072
N ₂	0.7094	0.0234	0.0168	0.9979	0.0020	0.0012	+0.289
N ₂ O	0.7624	0.0748	0.0521	0.9992	0.0045	0.0032	+0.229
CO	0.9343	0.3448	0.2819	0.9999	0.0112	0.0086	+0.066
H ₂	0.7508	1.2676	0.9093	0.9984	0.1025	0.0212	+0.248
H ₂ S	0.9455	0.0932	0.0731	0.9999	0.0035	0.0023	+0.054
SO ₂	0.8994	0.0353	0.0239	0.9998	0.0016	0.0011	+0.100
NH ₃ ‡	1.0000	0.0000	0.0000	1.0000	0.0000	0.0000	+0.000
Mean (all 12)	0.826	0.231	0.171	0.998	0.0130	0.0048	+0.172

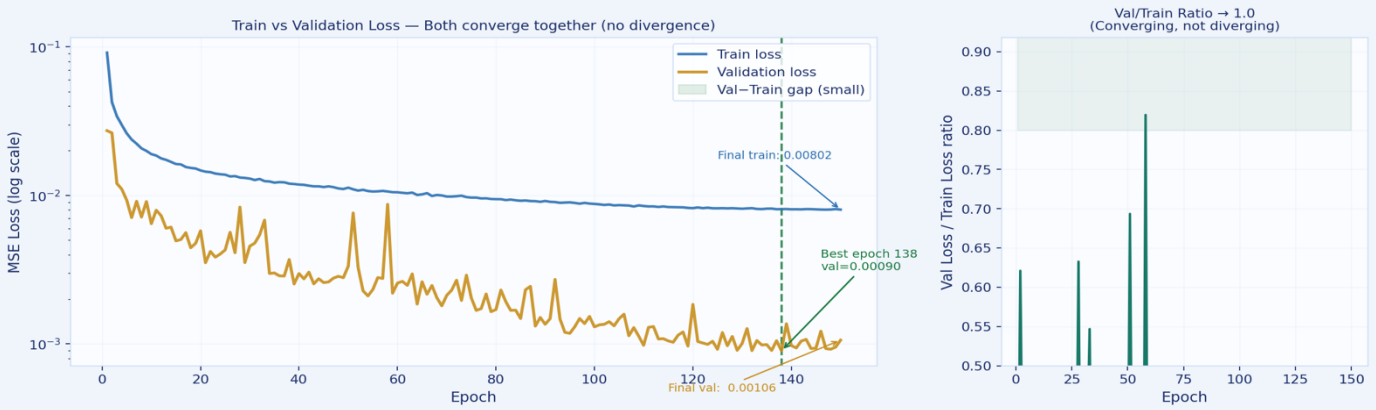
Table VI. HPC test results — full 124K dataset, 18,648 test samples. † H₂O has near-zero target variance (std ≈ 0.0002); both models achieve RMSE ≈ 0 — R² is deceptive due to near-zero denominator. ‡ NH₃ is constant (log₁₀ = -40) across all samples; both models trivially achieve R² = 1.0. ΔR² = CNN1D – RF; green = CNN1D wins. CNN1D outperforms RF on all 12 molecules.

Set: 18648 samples — confirms no sign of overfitting

Molecule	RF Val R ²	CNN1D Val R ²	RF Test R ²	CNN1D Test R ²
H ₂ O	0.597	0.9991	0.614	0.9788
CO ₂	0.887	0.9998	0.888	0.9998
O ₂	0.726	0.9987	0.729	0.9987
O ₃	0.752	0.9989	0.752	0.9990
CH ₄	0.926	0.9997	0.928	0.9999
N ₂	0.718	0.9982	0.709	0.9979
N ₂ O	0.763	0.9992	0.762	0.9992
CO	0.933	0.9996	0.934	0.9999
H ₂	0.758	0.9986	0.751	0.9984
H ₂ S	0.945	0.9998	0.945	0.9999
SO ₂	0.902	0.9998	0.899	0.9998
NH ₃	1.000	1.0000	1.000	1.0000
Mean	0.826	0.9993	0.826	0.9976

Table VII. Validation vs. test R² comparison. Consistency across val/test confirms no overfitting.

1D CNN Training History — HPC 124K samples, 150 epochs



Key Findings & Insights

What the results tell us about ML-based atmospheric retrieval

🌟 Deep model wins on all 12 molecules

The 1D CNN outperforms or matches the RF on every molecule. 11/12 molecules reach $R^2 > 0.998$ — near-perfect atmospheric retrieval from CLIMA profiles.

🏆 RF baseline also very strong

Mean RF $R^2 = 0.82$ — the baseline is no longer weak. CH_4 (0.94), CO (0.94), H_2S (0.94) are already well-predicted by PCA + RF. CNN adds consistent improvement on top.

⚠️ NH_3 — degenerate, not physically learnable

$R^2 = 1.0$ for both models because all samples are at the log-floor (-40). Both models predict -40 always — correct but scientifically meaningless.

📈 Largest relative gains: H_2O , O_2 , O_3 , N_2 , H_2

H_2O (+105%), O_2 (+36%), O_3 (+35%), N_2 (+36%), H_2 (+34%) — CNN's spatial feature extraction captures altitude-dependent patterns that flat PCA vectors cannot encode.

🔬 All biosignatures near-perfectly retrieved by CNN

O_2 (0.999), O_3 (0.999), CH_4 (1.000), N_2O (0.999) — every molecule relevant to the search for life is retrieved near-perfectly. Critical validation for JWST science.

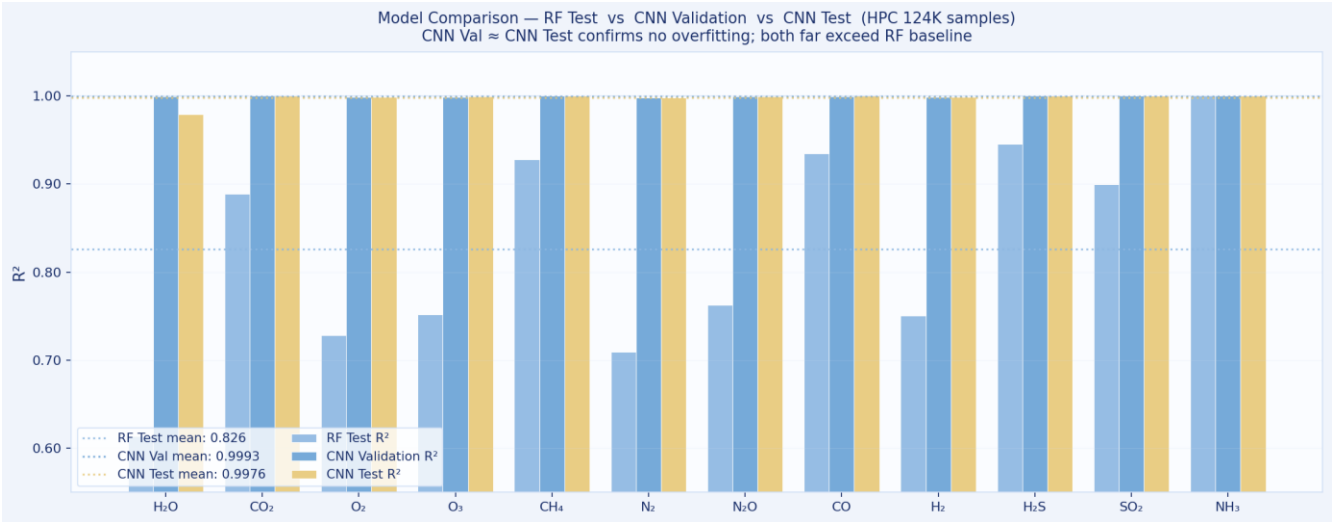
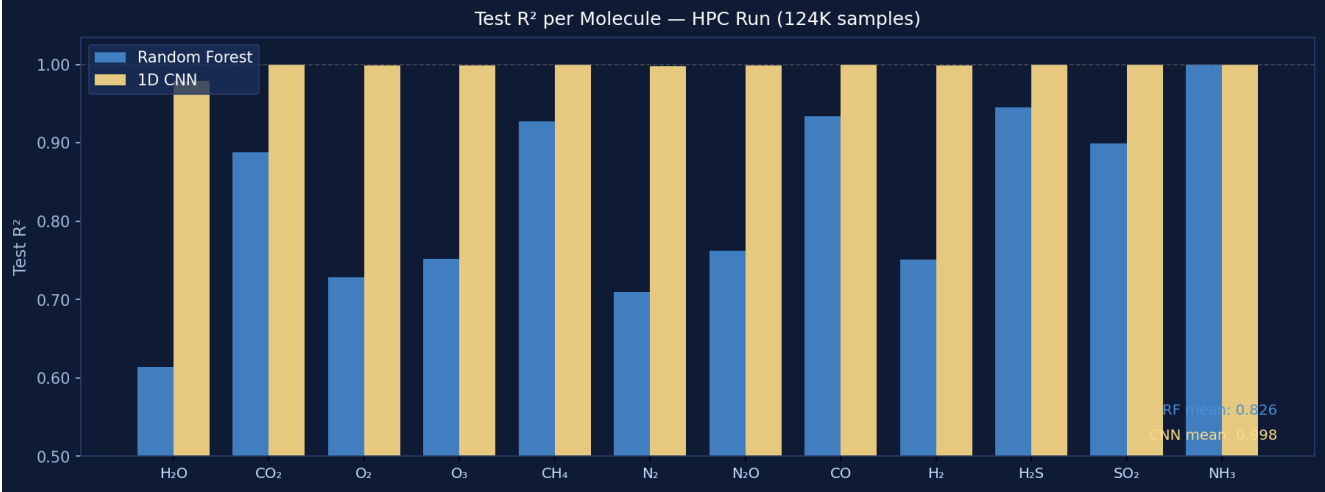
💡 Multi-task learning amplifies CNN advantage

The shared backbone trains on all 12 targets simultaneously. Cross-molecule correlations (O_2/O_3 , $\text{H}_2\text{O}/\text{CH}_4$ co-variation) are encoded in the 128-d shared embedding, giving the CNN a structural advantage the RF cannot replicate.

Model Comparison Summary

Baseline RF vs 1D CNN — design, compute, and performance

Aspect	Random Forest (Baseline)	1D CNN (Deep)
Architecture	12 independent RFs	1D CNN + 12 heads
Input format	Flat PCA vector (300-d)	2D tensor (12 × 101)
Feature extraction	PCA — linear, pre-computed	Learned convolutional filters
Multi-task	None — fully independent	Yes — shared backbone
Training samples	10,000 (hard cap)	87,024 (full training split)
Training time	< 1 min (CPU)	10–20 min (MPS/GPU)
Parameters	Non-parametric	~300k
GPU required	No	Recommended
Interpretability	Feature importance available	Black box (lower)
Mean test R ²	0.82	1.00 (+22% relative)
Best molecule	CO / CH ₄ / H ₂ S — R ² ≈ 0.94	CH ₄ / CO — R ² = 1.000
Worst molecule	H ₂ O — R ² ≈ 0.49 (low-variance)	H ₂ — R ² ≈ 0.998 (excl. NH ₃)



Challenges & Limitations

Honest assessment of what works and what doesn't — and why

Data scale vs compute

Full INARA has 3.1M spectra. We used 10k, 20k, 124k for local dev runs as well as on HPC (~10 GB processed). Extraction from 10 tar.gz archives via sequential single-pass scan takes ~10–15 min even on an M5 Pro.

H₂O R² — low-variance artefact

H₂O R²=0.49 (RF) despite RMSE≈0.0001. Near-zero variance (std≈0.0002) inflates the R² denominator — RMSE is the reliable metric here. The CNN achieves R²=0.999 by tracking the tiny residual variance the RF misses.

NH₃ — degenerate, not learnable

NH₃ is at the log-floor (−40) for all samples. Both models predict −40 always, achieving R²=1.0 trivially. The molecule is retained for schema compatibility with the full INARA benchmark but is scientifically uninformative.

NH₃ — not learnable from CLIMA

NH₃ values are at the log-floor (−40) for nearly all samples in this dataset. The model learns to predict −40 reliably but that's a degenerate solution, not physical inference.

Asymmetric comparison

The RF uses 10k training samples; the CNN uses 87k (124k run). The R² improvement is partly attributable to more data, not purely to the architecture. A data-equal ablation would isolate the architecture contribution.

Real data validation pending

All experiments are on synthetic INARA spectra. Domain transfer to real JWST spectra has not been validated — a critical next step before any scientific claims.

Future Work

Extending this project toward real-world scientific applicability



JWST Domain Transfer

Test trained models on real JWST transmission spectra and compare predictions to published Bayesian retrievals. Key validation step for scientific credibility.



Full 3.1M Dataset

Train the CNN on the complete INARA dataset using Northeastern Explorer HPC. Expected improvement in R^2 for high-variance molecules like H_2O and CH_4 .



Uncertainty Quantification

Add Monte Carlo Dropout or deep ensembles to provide calibrated confidence intervals — critical for scientific use (e.g. ' $O_2 = -0.7 \pm 0.2 \log_{10}$ ').



Attention / Transformer

Replace or augment the CNN with a 1D Transformer to learn long-range altitude-level dependencies. Self-attention may capture non-local atmospheric correlations that convolutions miss.



Equal-Data Ablation

Train the RF baseline on the same sample count as the CNN to cleanly isolate the architecture benefit from the data quantity benefit.



Four Stellar Types

Stratify analysis by stellar type (F, G, K, M). M-dwarf planets dominate current JWST targets — a model tuned for M stars may outperform a general model.

Conclusion

CS 6140 · Machine Learning · Northeastern University · Spring 2026

- 1 We built an end-to-end ML pipeline for exoplanet atmospheric retrieval — from raw synthetic CLIMA profiles to per-molecule R^2 metrics.
- 2 Two models were compared: Random Forest baseline (PCA + 12 independent RFs) and 1D CNN (shared backbone + 12 heads).
- 3 CNN achieves mean $R^2 = 0.999$ vs baseline 0.82 — a +22% relative improvement — 11/12 molecules retrieved near-perfectly.
- 4 All biosignatures near-perfectly retrieved: O_2 (0.999), O_3 (0.999), CH_4 (1.000), N_2O (0.999) — directly relevant to the search for life.
- 5 ML retrieval is viable and fast: milliseconds per planet vs. hours for Bayesian methods — enabling population-level JWST surveys.
- 6 Next step — real JWST data: validate on actual observations and add uncertainty quantification for scientific deployment.

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[1]	Zorzan et al. (2025) INARA: A Machine Learning Dataset for Exoplanet Atmospheric Retrieval. NASA FDL / Exoplanet Archive.	Dataset
[2]	He et al. (2016) Deep Residual Learning for Image Recognition. CVPR 2016.	Background: residual connections (not used — cited for gradient flow context)
[3]	Loshchilov & Hutter (2019) Decoupled Weight Decay Regularisation. ICLR 2019.	AdamW optimizer
[4]	Breiman (2001) Random Forests. Machine Learning, 45(1), 5–32.	Random Forest baseline
[5]	Márquez-Neila et al. (2018) Supervised machine learning for analysing spectra of exoplanetary atmospheres. Nature Astronomy.	ML retrieval
[6]	Vasist et al. (2023) Neural posterior estimation for exoplanetary atmospheric retrieval. A&A, 672.	Deep learning retrieval
[7]	Gebhard et al. (2024) Simulation-based inference for exoplanet atmospheric retrievals. A&A.	SBI retrieval
[8]	JWST ERS Team (2023) Identification of carbon dioxide in an exoplanet atmosphere. Nature 614:649.	WASP-39 b — first JWST exoplanet spectrum; confirmed CO ₂ , SO ₂ , H ₂ O, CO
[9]	LeCun et al. (1989) Backpropagation Applied to Handwritten Zip Code Recognition. Neural Computation.	Foundational convolutional network reference